SQL Assignment – Departments Table

Operations

Problem Statement

Create a table named Departments with the following columns:

• dept_id

• name

• count_emp_dept

• dept_mail_id

Insert at least 10–15 records into the table.

Perform the following SQL operations on the table:

• CREATE

• INSERT

• SELECT

• WHERE

• ORDER BY

• UPDATE

• DELETE

• ALTER

• TRUNCATE

• DROP

• Logical operators (AND, OR)

• Comparison operators (=, >, <, >=, <=, !=)

1. Create Table

```
CREATE TABLE Departments (

dept_id INT PRIMARY KEY,
```

name VARCHAR(50),

count_emp_dept INT,

);

dept_mail_id VARCHAR(100)

2. Insert Records (15 Rows)

INSERT INTO Departments VALUES

(1, 'HR', 15, 'hr@company.com'),

(2, 'Finance', 20, 'finance@company.com'),

(3, 'IT', 50, 'it@company.com'),

(4, 'Marketing', 25, 'marketing@company.com'),

(5, 'Sales', 40, 'sales@company.com'),

(6, 'Admin', 10, 'admin@company.com'),

(7, 'Operations', 35, 'operations@company.com'),

(8, 'Legal', 8, 'legal@company.com'),

(9, 'R&D', 30, 'rnd@company.com'),

(10, 'Support', 22, 'support@company.com'),

(11, 'Procurement', 12, 'procurement@company.com'),

(12, 'Logistics', 18, 'logistics@company.com'),

(13, 'Security', 14, 'security@company.com'),

(14, 'Design', 16, 'design@company.com'),

(15, 'QA', 28, 'qa@company.com');

3. SELECT Command

Select all records

SELECT * FROM Departments;

Select specific columns

SELECT name, count_emp_dept FROM Departments;

4. WHERE Clause with Comparison Operators

Departments with more than 20 employees

SELECT * FROM Departments

WHERE count_emp_dept > 20;

Departments with exactly 10 employees

SELECT * FROM Departments

WHERE count_emp_dept = 10;

Departments with employee count not equal to 15

SELECT * FROM Departments

WHERE count_emp_dept != 15;

5. Logical Operators (AND, OR)

Using AND

SELECT * FROM Departments

WHERE count_emp_dept > 20 AND count_emp_dept < 40;

Using OR

SELECT * FROM Departments

WHERE name = 'HR' OR name = 'Finance';

6. ORDER BY

Order by employee count (ascending)

SELECT * FROM Departments

ORDER BY count_emp_dept ASC;

Order by employee count (descending)

SELECT * FROM Departments

ORDER BY count_emp_dept DESC;

7. UPDATE Command

Increase employee count of HR department

```
UPDATE Departments
SET count_emp_dept = 18
WHERE dept_id = 1;
```

8. DELETE Command

Delete department with dept_id = 8

```
DELETE FROM Departments
WHERE dept_id = 8;
```

9. ALTER Command

Add a new column (location)

```
ALTER TABLE Departments
ADD location VARCHAR(50);
```

Modify column size

```
ALTER TABLE Departments
MODIFY name VARCHAR(100);
```

10. TRUNCATE Command

Removes all records but keeps table structure.

```
TRUNCATE TABLE Departments;
```

11. DROP Command

Deletes entire table including structure.

```
DROP TABLE Departments;
```